Appendix A - Beagle Marine Park Natural Values Report (stereo BRUV)

South-east Marine Parks Network

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Natural Values

Understanding of natural values

Benthic ecosystem extent

Observations

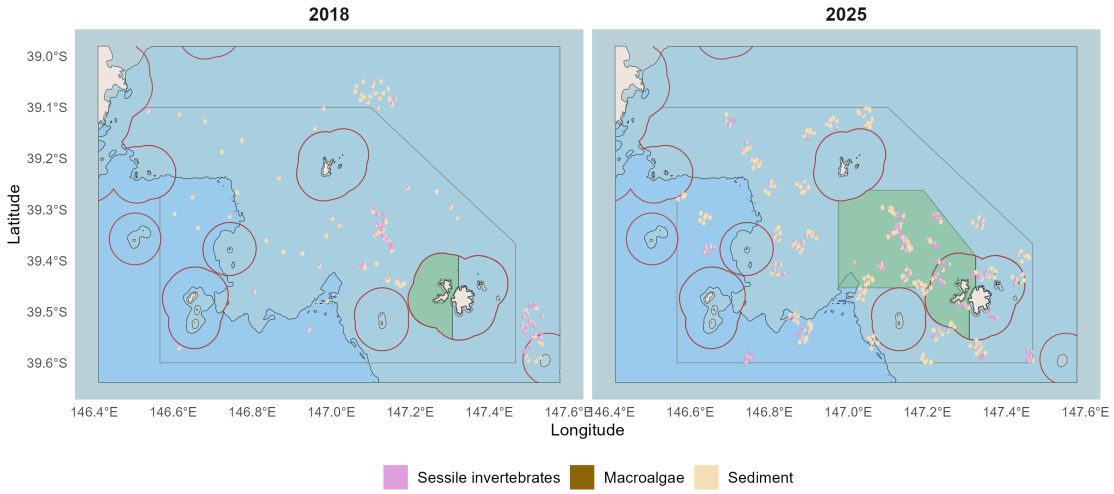


Figure A.1: Habitat classes observed in spatially balanced stereo-BRUV imagery visualised as spatial pie charts. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contours at 30 and 70 m are shown.

Dominant habitat types

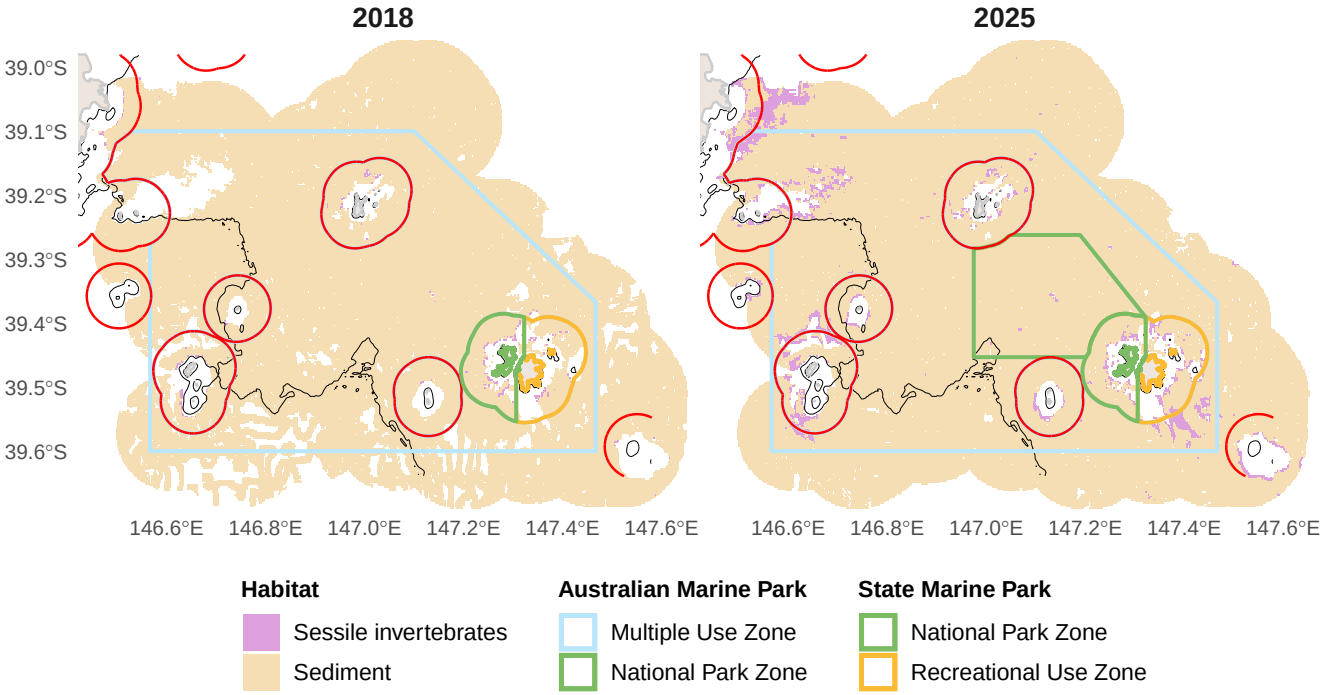


Figure A.2: Predicted dominant habitat type from stereo-BRUV groundtruthing. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contours at 30 and 70 m are shown.

Habitat distribution and probability

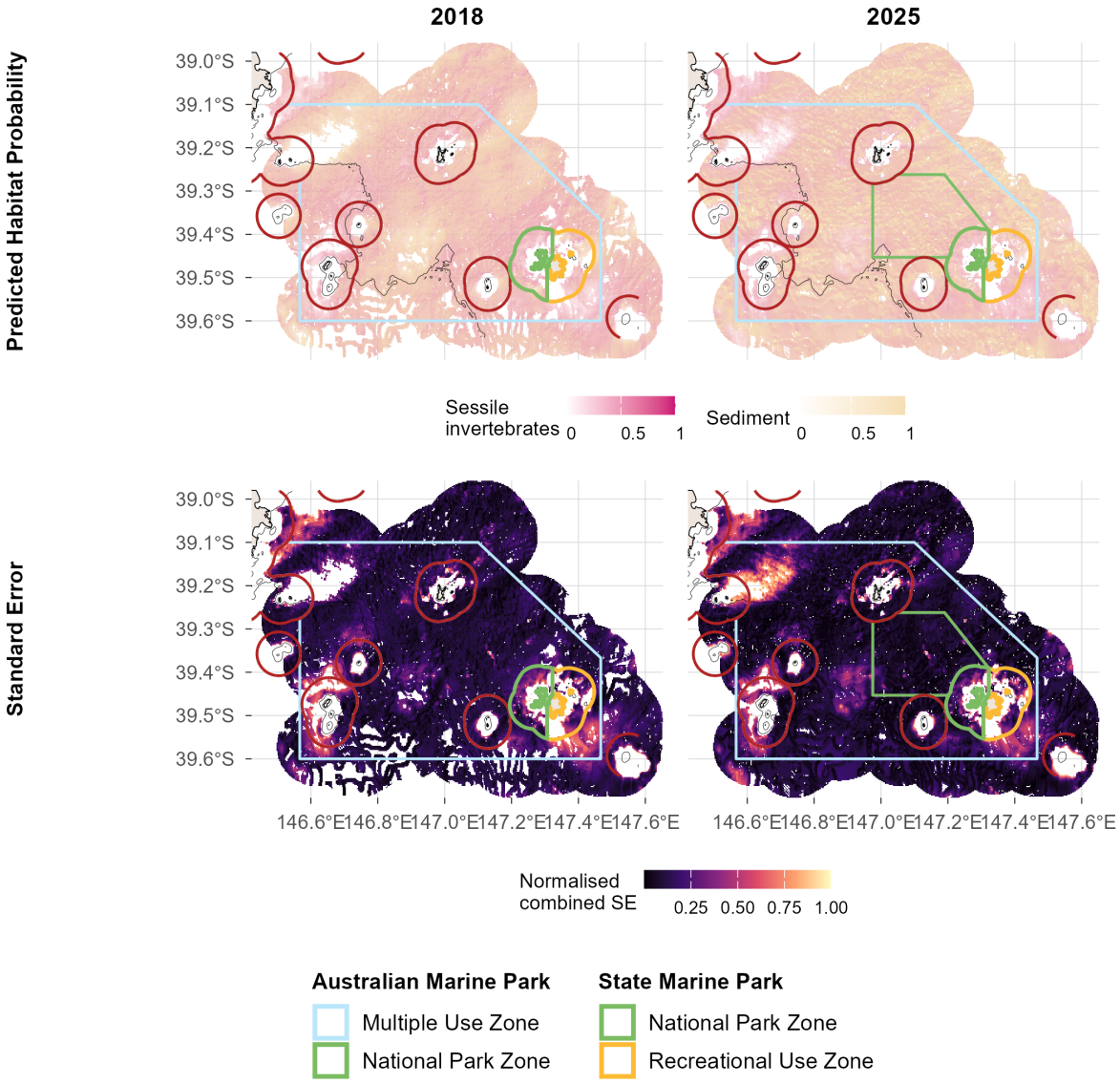


Figure A.3: Predicted habitat distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contours at 30 and 70 m shown.

Benchmarks of benthic natural values

Sediment

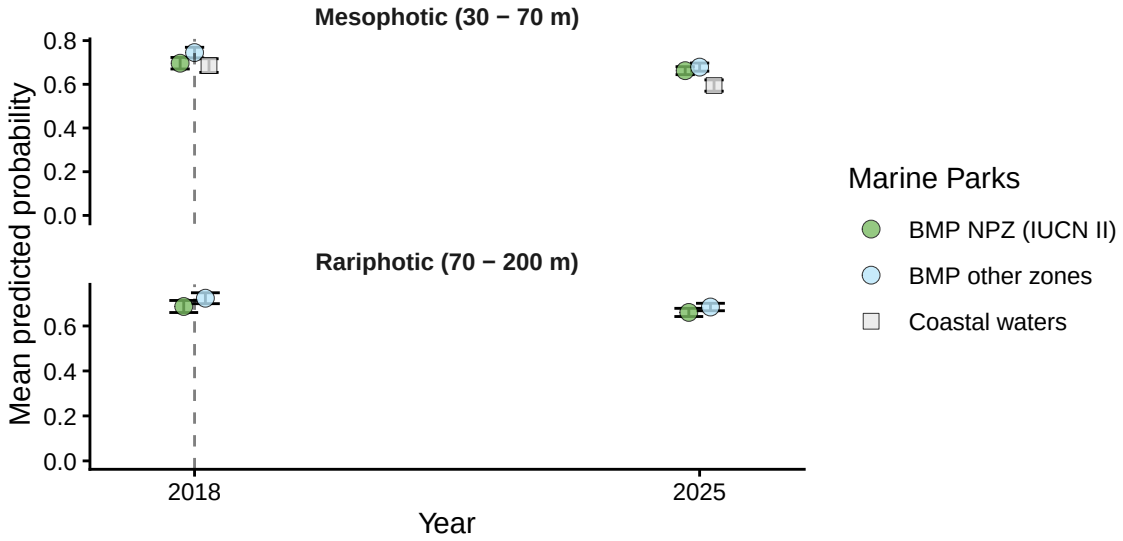


Figure 4.1: Sediment: temporal plots by depth contour from stereo-BRUV groundtruthing. BMP = Beagle Marine Park (Commonwealth).

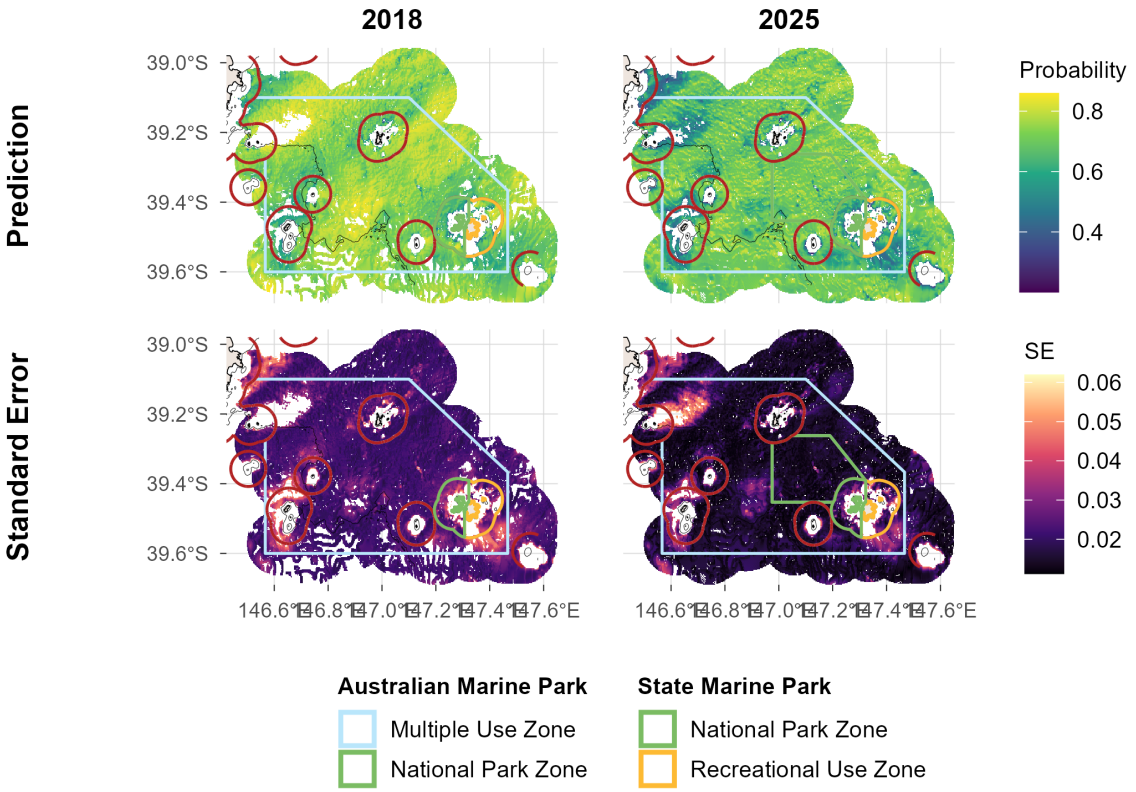


Figure 4.2: Sediment: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contours at 30 and 70 m are shown.

Sessile invertebrates

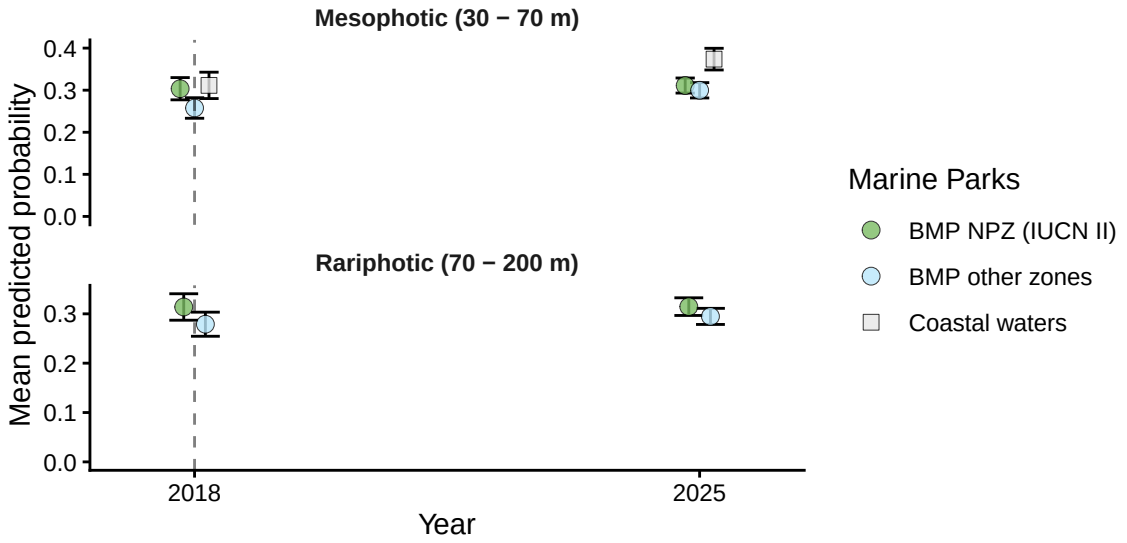


Figure 5.1: Sessile invertebrates: temporal plots by depth contour from stereo-BRUV groundtruthing. BMP = Beagle Marine Park (Commonwealth).

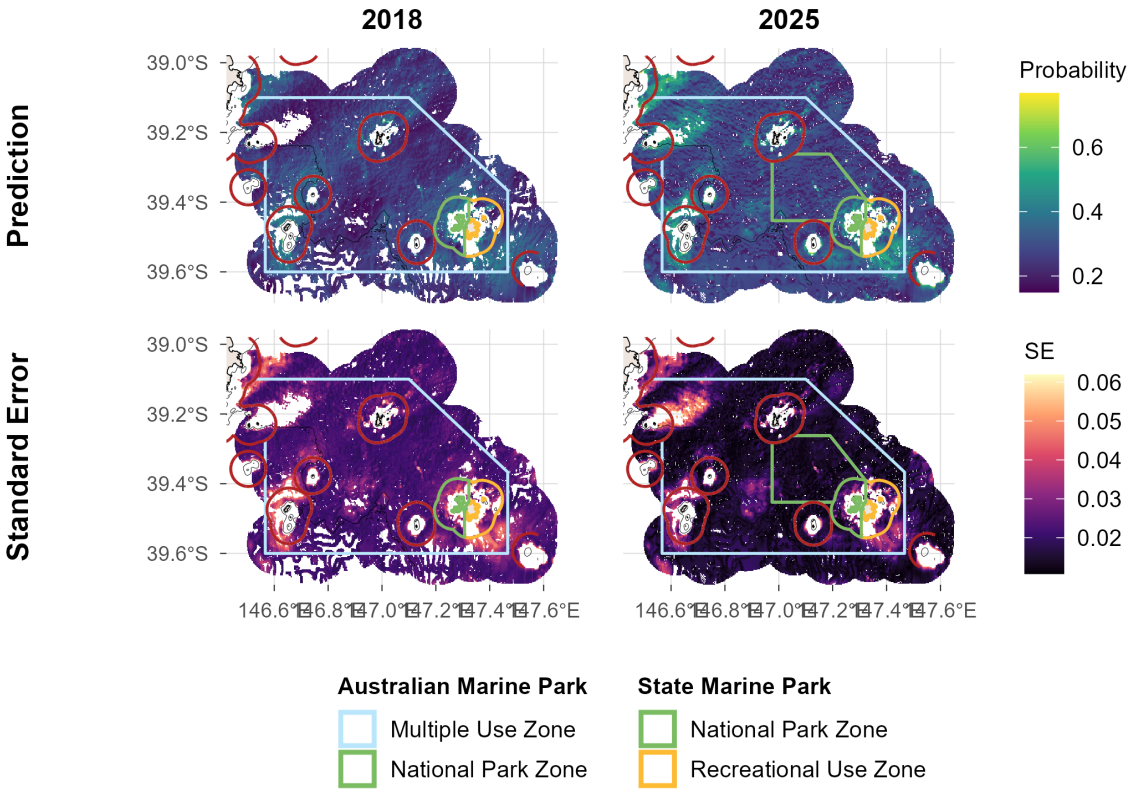


Figure 5.2: Sessile invertebrates: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contours at 30 and 70 m are shown.

Benchmarks of demersal fish natural values

Survey effort

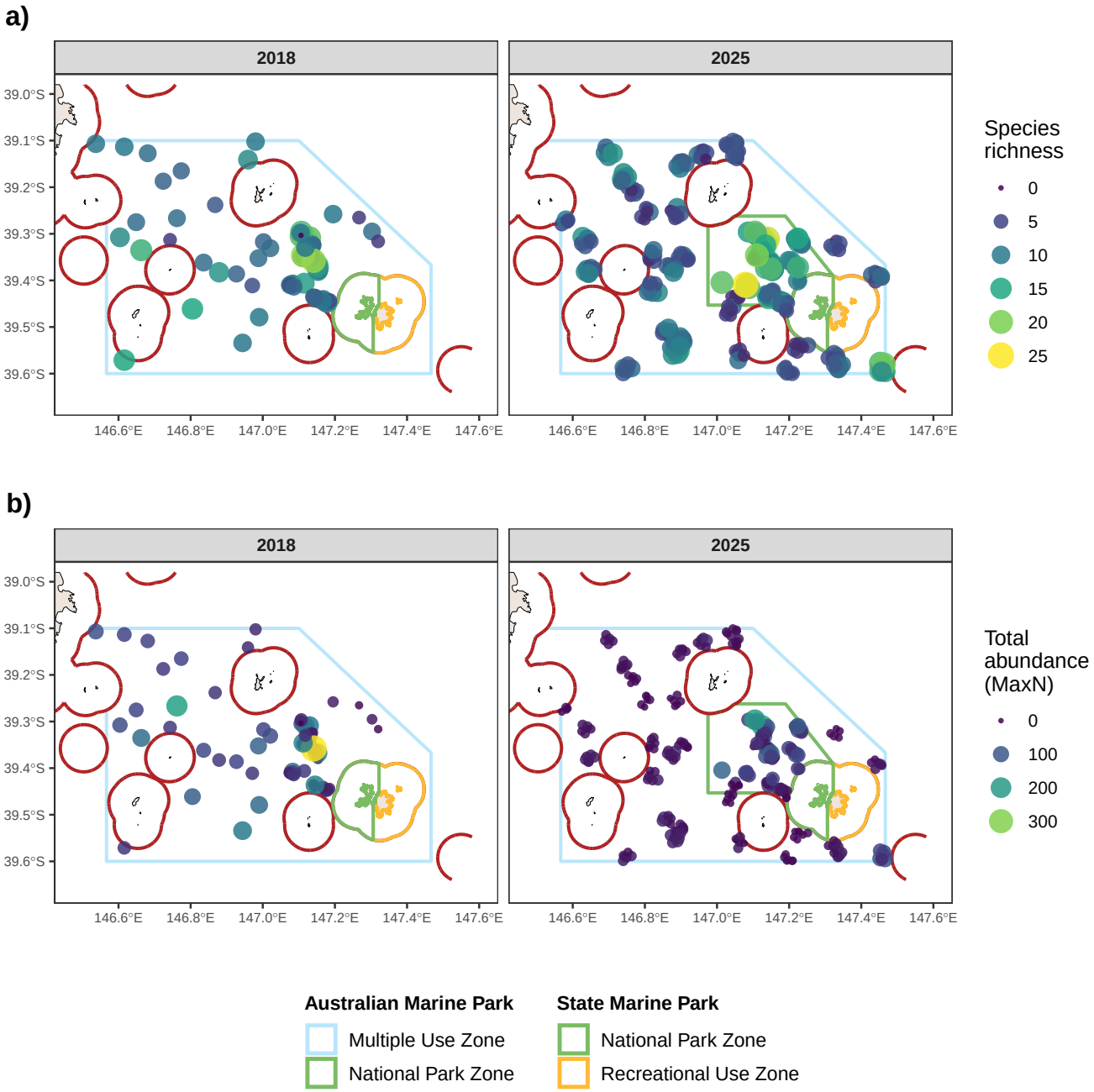


Figure 6: Spatial distribution of observed species richness and total abundance (MaxN) per stereo-BRUV drop, pooled across all deployments. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters.

Species richness

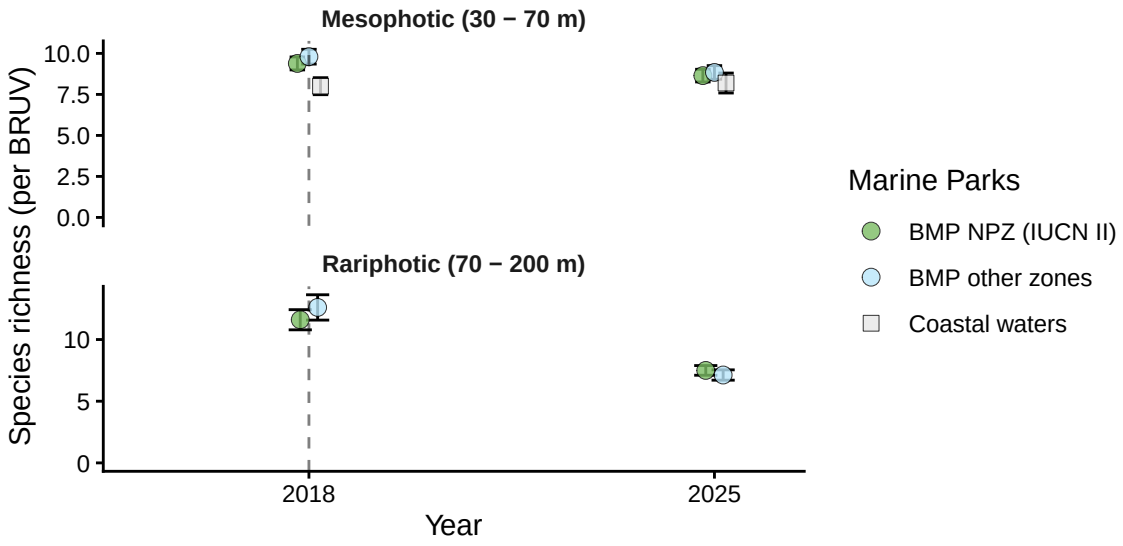


Figure 7.1: Demersal fish - Species richness: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. BMP = Beagle Marine Park (Commonwealth).

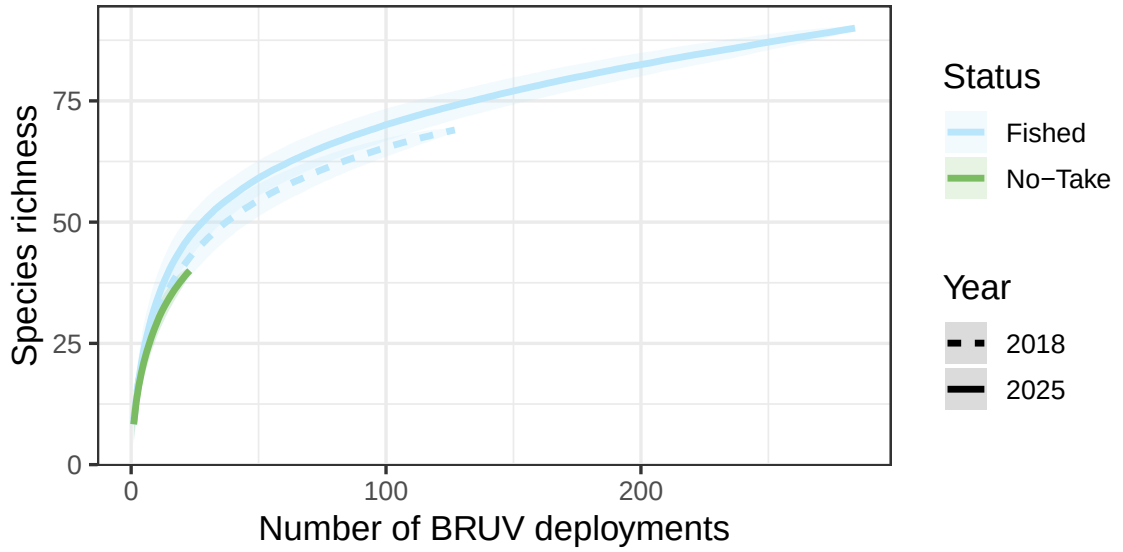


Figure 7.2: Demersal fish - Species richness: species accumulation curve by year from stereo-BRUV groundtruthing. No-Take deployments in this curve are from State marine parks.

Species richness

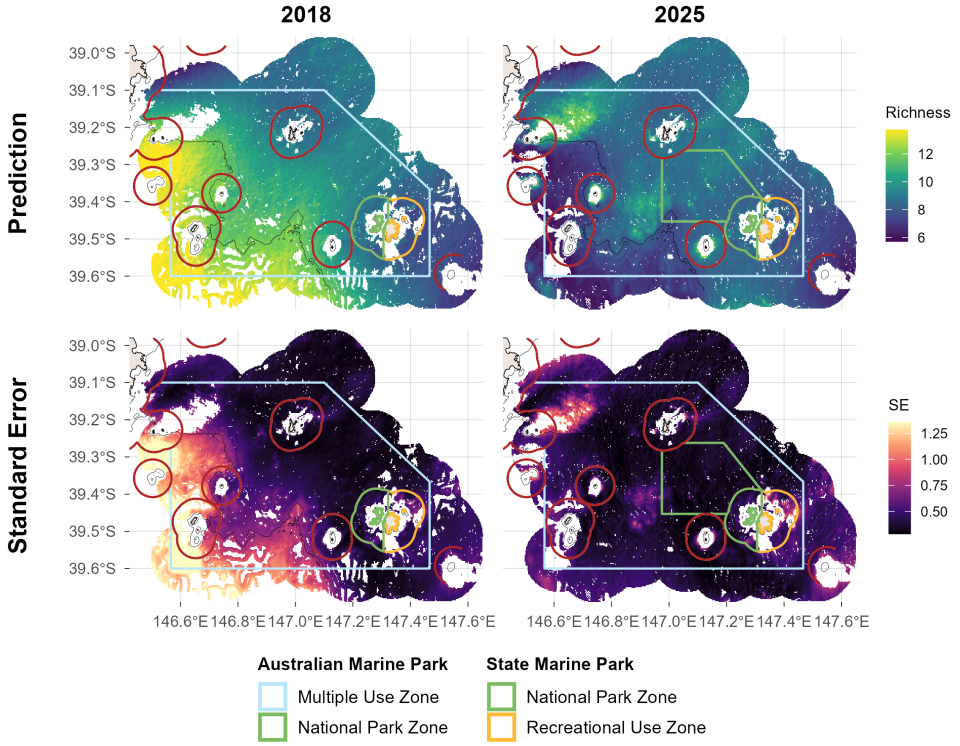


Figure 7.3: Demersal fish - Species richness: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contours at 30 and 70 m are shown.

Total abundance

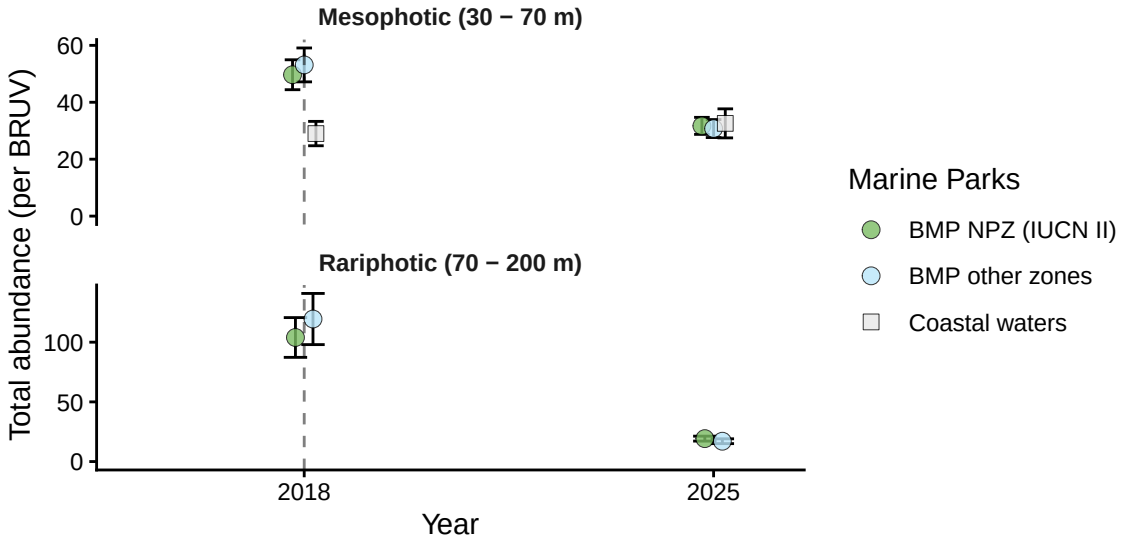


Figure 8.1: Demersal fish - Total abundance: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. BMP = Beagle Marine Park (Commonwealth).

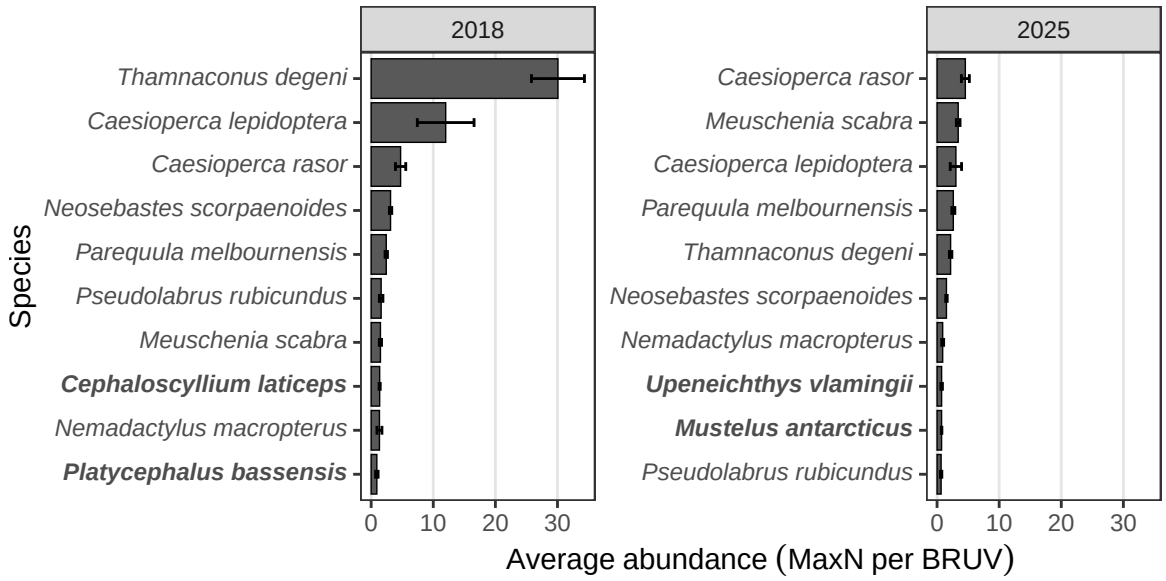


Figure 8.2: Demersal fish - Total abundance: top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing. **Bold** species are unique within the 10 greatest biomass species by year.

Total abundance

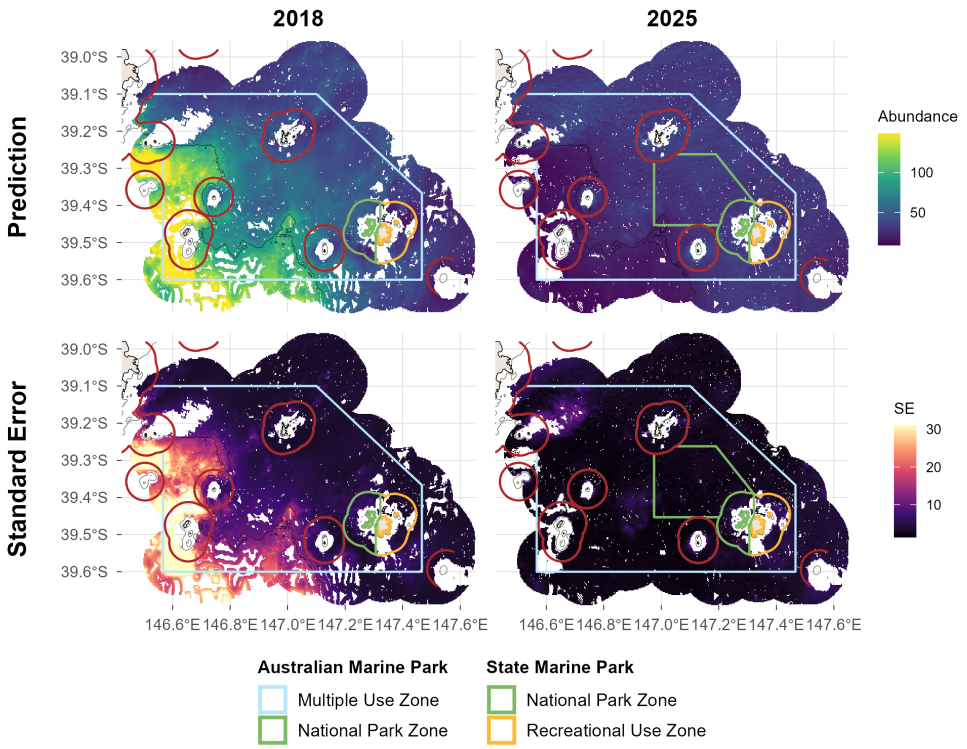


Figure 8.3: Demersal fish - Total abundance: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contours at 30 and 70 m are shown.

Large Reef Fish Index*

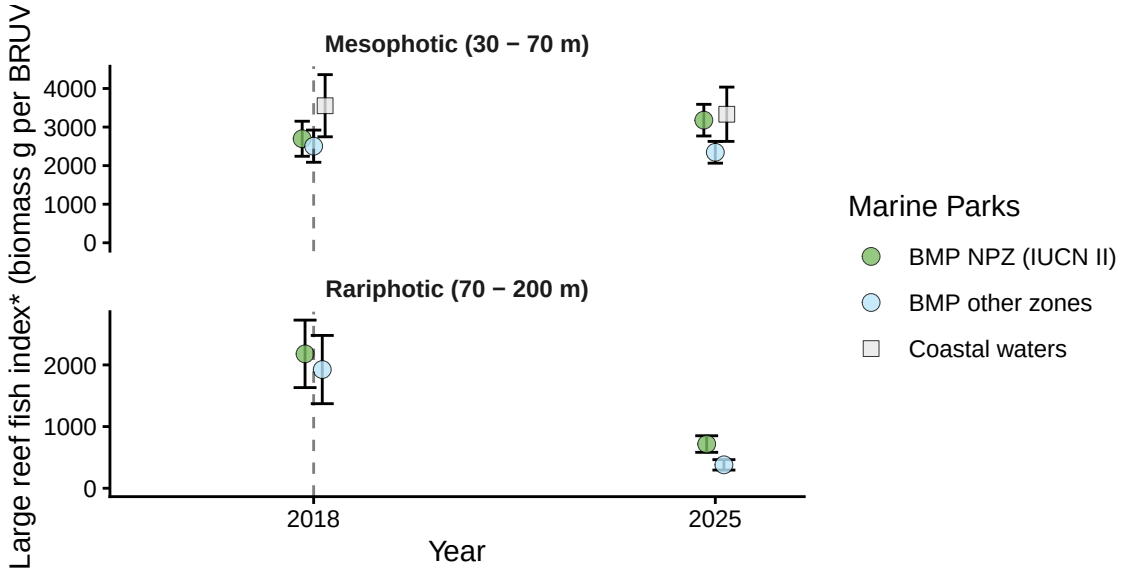


Figure 9.1: Demersal fish - Large Reef Fish Index*: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. BMP = Beagle Marine Park (Commonwealth). *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

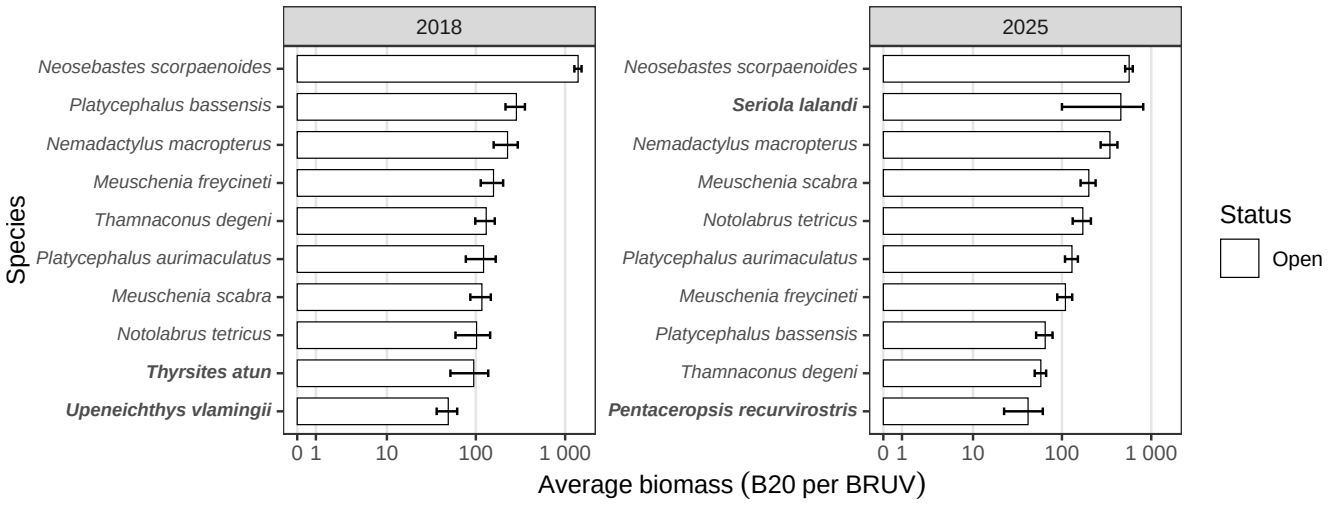


Figure 9.2: Demersal fish - Large Reef Fish Index*: top 10 greatest biomass species (pooled across deployments) from stereo-BRUV groundtruthing. **Bold** species are unique within the 10 greatest biomass species by year.

Large Reef Fish Index*

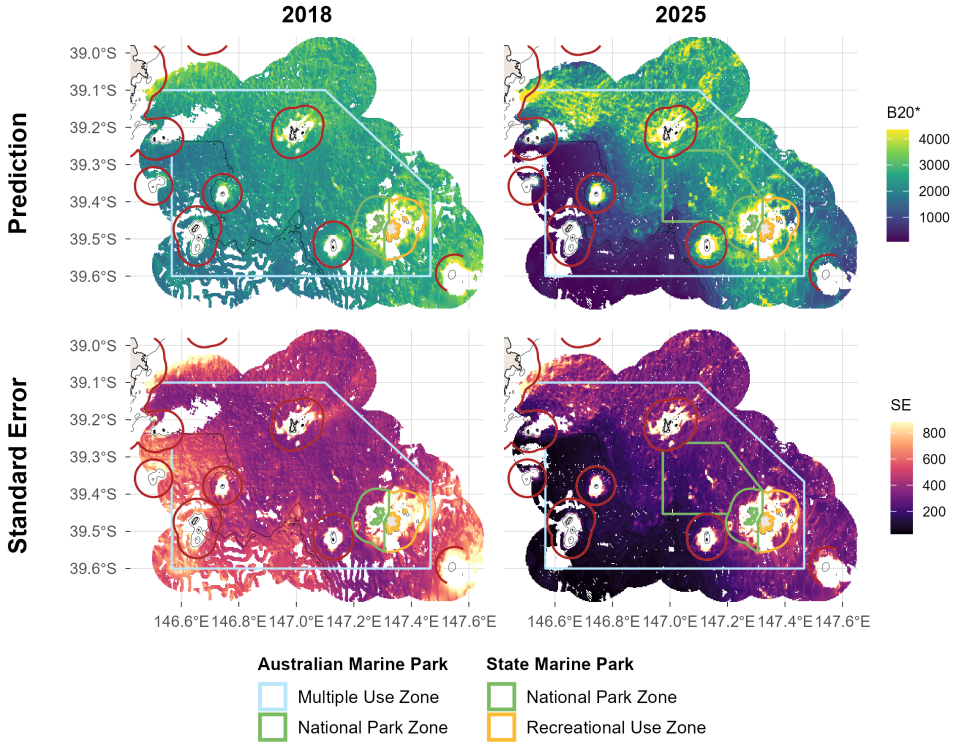


Figure 9.3: Demersal fish - Large Reef Fish Index*: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contours at 30 and 70 m are shown.

Reef fish thermal index

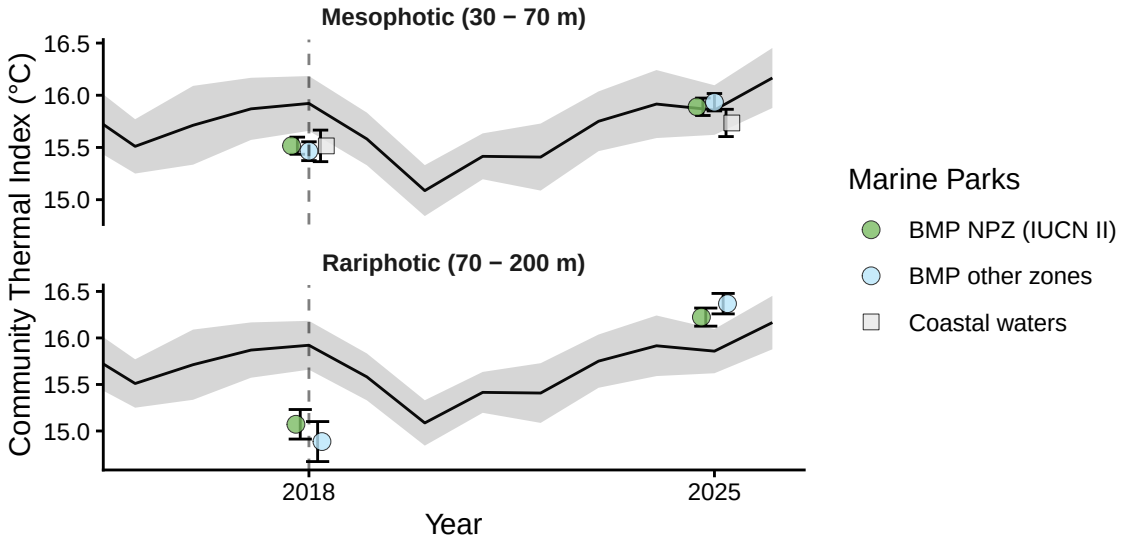


Figure 10.1: Demersal fish - Reef Fish Thermal Index: benchmark plots by zone and depth contour, with monthly sea surface temperature (black line, grey band = SD) overlaid, from stereo-BRUV groundtruthing. BMP = Beagle Marine Park (Commonwealth).

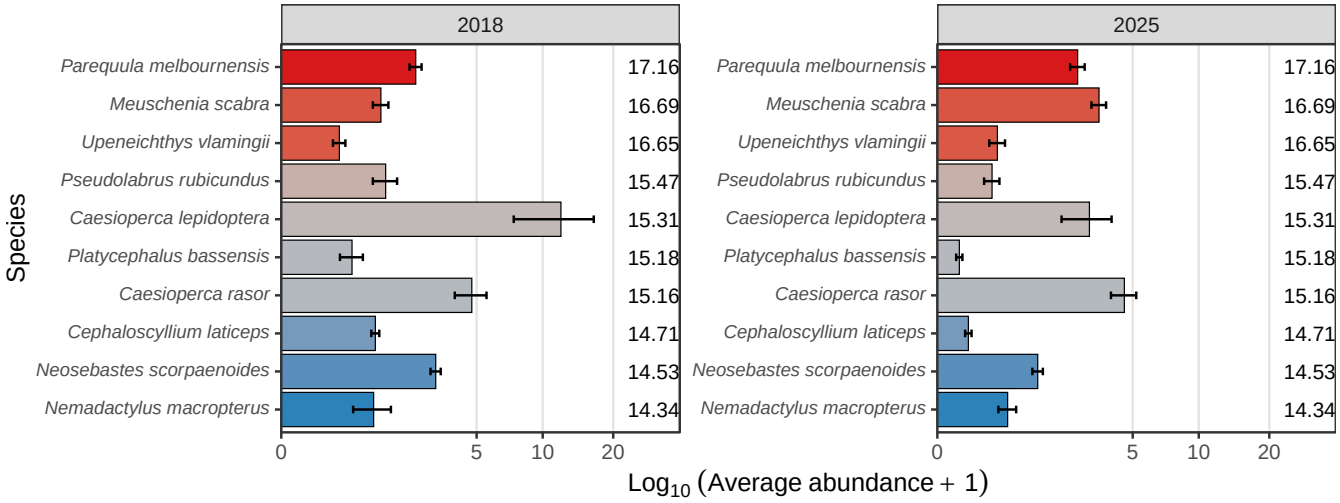


Figure 10.2: Demersal fish - Reef Fish Thermal Index: species thermal niche of top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing. **Bold** species are unique within the 10 greatest biomass species by year.

Reef fish thermal index

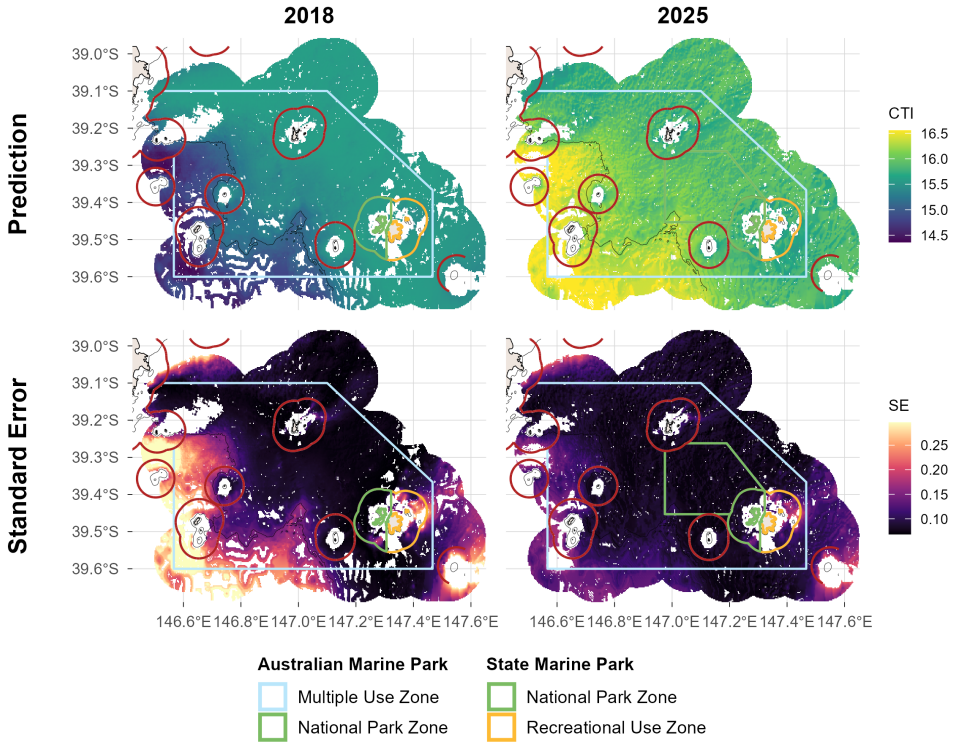


Figure 10.3: Demersal fish - Reef Fish Thermal Index: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contours at 30 and 70 m are shown.

Threatened species

Table A.1: Threatened species recorded in stereo-BRUV imagery across all survey years, showing IUCN Red List category, the number of samples each species was recorded in, and the total abundance (MaxN) summed across those samples. *This species is also listed as 'Conservation dependent' on the EPBC Act List of Threatened Fauna.

Species	Common name	Family	IUCN category	Samples	Total MaxN	Years recorded
Sepia apama	Giant cuttlefish	Sepiidae	Near Threatened	41	46	2018, 2025
Notorynchus cepedianus	Broadnose Shark	Hexanchidae	Vulnerable	25	25	2018, 2025
Spiniraja whitleyi	Melbourne Skate	Rajidae	Vulnerable	24	24	2018, 2025
*Galeorhinus galeus**	School Shark	Triakidae	Critically Endangered	23	24	2025
Carcharhinus brachyurus	Bronze Whaler	Carcharhinidae	Vulnerable	1	1	2025
Trygonoptera testacea	Common Stingaree	Urolophidae	Near Threatened	1	1	2018

Threatened shark and ray species

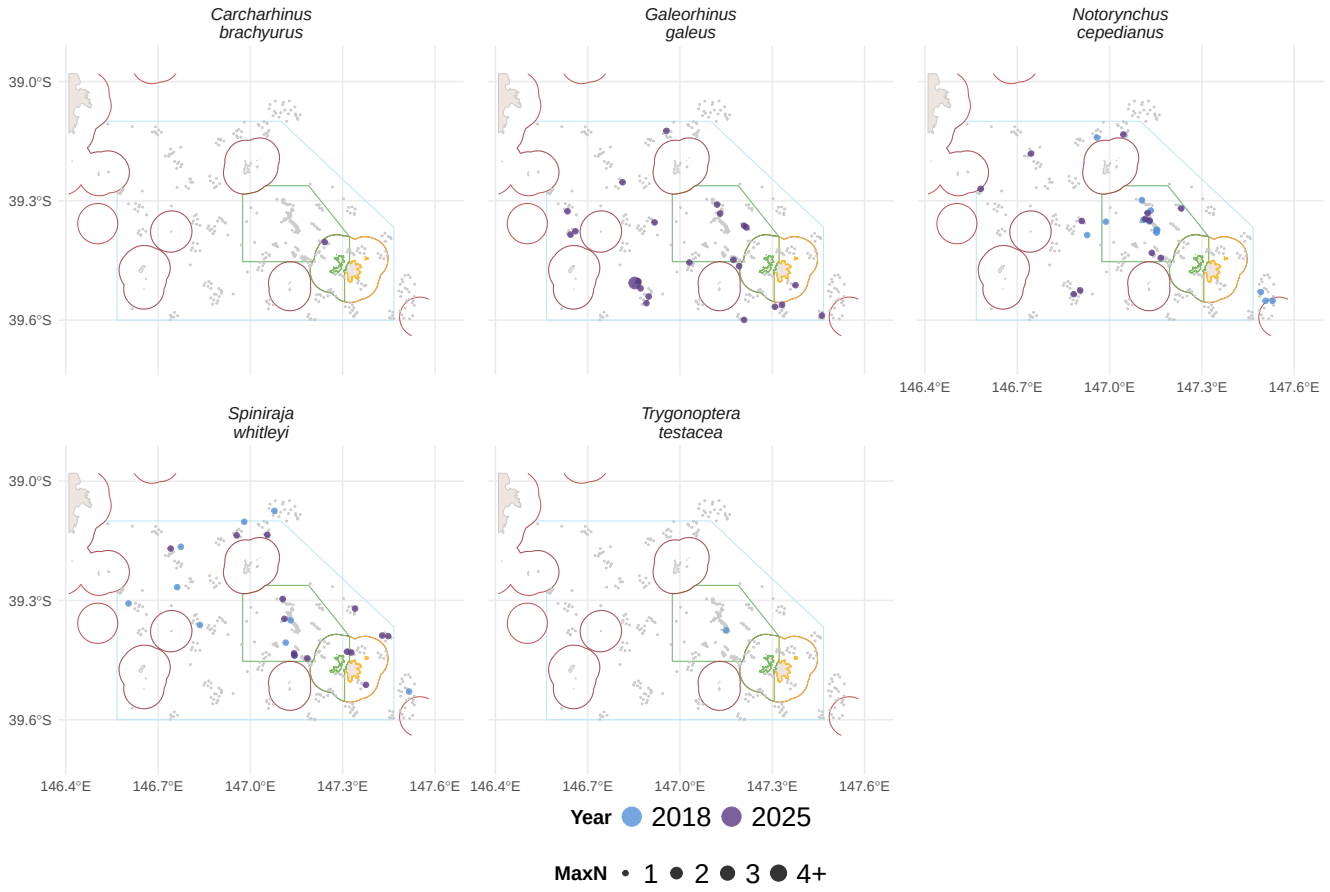


Figure 11.1: Occurrence of threatened, IUCN Red List category, sharks and rays in stereo-BRUV imagery. Points are scaled by abundance (MaxN) and coloured by survey year; grey points show every stereo-BRUV deployment. State and Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters.

Other shark and ray species

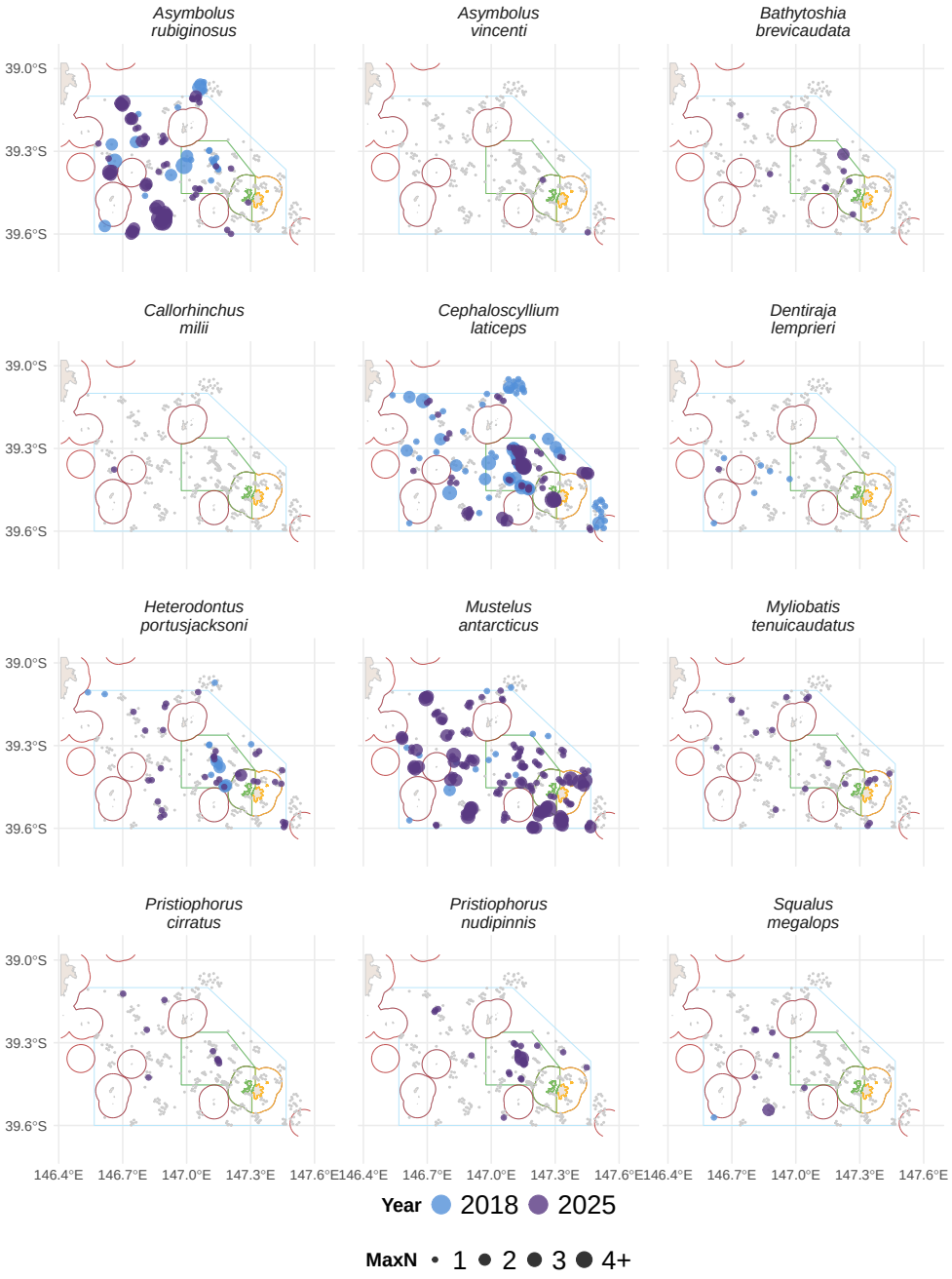


Figure 11.2: Occurrence of shark and ray species that are not threatened in stereo-BRUV imagery. Points are scaled by abundance (MaxN) and coloured by survey year; grey points show every stereo-BRUV deployment, so a panel with no coloured points is a genuine absence rather than an absence of sampling. State and Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters.

Other shark and ray species

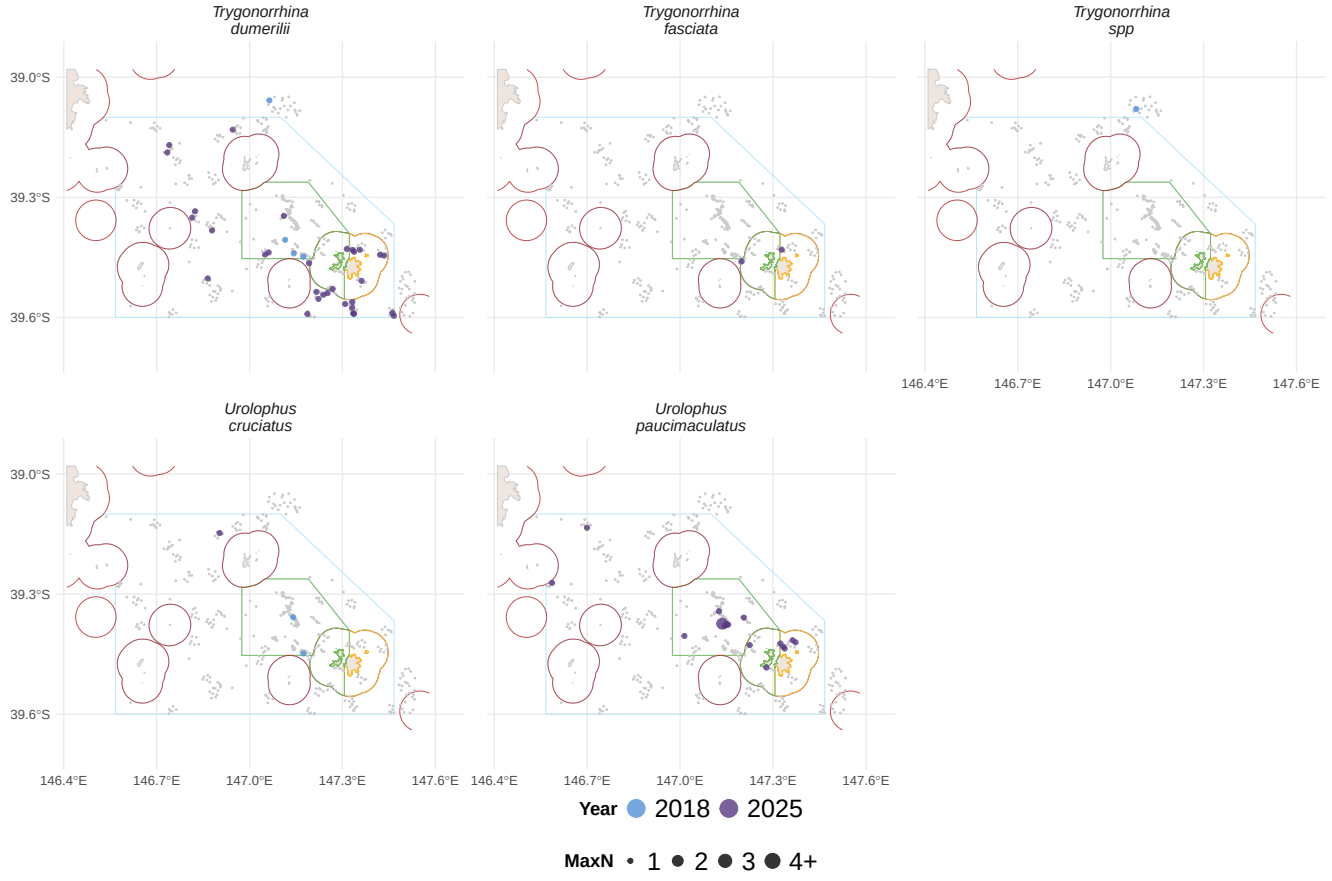


Figure 11.3: Occurrence of shark and ray species that are not threatened in stereo-BRUV imagery (continued). Points are scaled by abundance (MaxN) and coloured by survey year; grey points show every stereo-BRUV deployment, so a panel with no coloured points is a genuine absence rather than an absence of sampling. State and Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters.